Marc Jakobi

Renewable Energy Systems | Software Engineer

*«Insanity is running the same build over and over again
and getting different results»*

Experienced software engineer and open source maintainer combining **renewable energy systems expertise** with deep experience in **distributed systems** & **functional programming**. Proven track record in **building reliable backend platforms**, shaping clean architectures, and taking ownership of quality-focused delivery through **test automation, CI/CD, reproducible builds, and collaborative XP practices**.

Professional skills

Languages	Haskell, Nix, Rust, Elixir, C, Scala, Kotlin, Java, Lua, Python, C++, TypeScript, MATLAB
Software engineering	Functional software architecture, TDD, BDD, property testing, pair programming, distributed systems, CI/CD, quality-focused automation
Renewable energy	Photovoltaics, batteries, e-mobility, thermal systems, sector coupling, energy management, electric grid
Dev environment	Vim/Neovim, Linux (NixOS), tmux, Nushell, jj-vcs *UNIX is my IDE*
Simulation	Polysun, Simulink, TRNSYS
Scientific documentation	LaTeX, Typst, Pandoc

Professional career

02/2022–02/2026 **Backend developer**, *tiko Energy Solutions AG*

Architected and developed distributed systems primarily in Haskell, leveraging Nix for reproducible builds. (80 % workload since 2023)

Virtual Power Plant and IoT devices

- Microservices written in Haskell for high-throughput, low-latency device management and ancillary services (FCR + aFRR)
- Redis/Valkey, Kafka, CBOR, MQTT, PostgreSQL, Hazelcast, GraphQL

Data pipelines

- Resilient event-driven pipelines written in Haskell
- MQTT, Kafka, Thrift, Protobuf, Avro, rocksDB-cloud, TimescaleDB, RabbitMQ

Web services

- Built and maintained scalable Haskell APIs (servant, wai/warp)

Quality & test automation

- Propagated behaviour driven development practices, increasing integration test coverage.
- Designed KVM-based integration tests of distributed systems using the nixosTest framework, enabling zero rollbacks in VPP service deployments.
- Led efforts to improve the build system, organise and unify the codebase, write and update documentation, and keep dependencies up to date.

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Continuous integration, deployment
- Hydra, Morph, Colmena, GitLab, k8s, ArgoCD, Kustomize, Terraform, AWS, SOPS, Dhall
- Release management, regular system deployments

API architecture and collaboration
- Worked with Data Science, Full Stack and Firmware teams to design cross-system APIs.
- Mentored, supported and pair-programmed with junior backend team members.

Monitoring, Alerting
- Implemented OpenTelemetry instrumentation, enabling observability across services.
- Prometheus, Grafana, Zabbix, PagerDuty, Elasticsearch, Kibana
- On-call rotation, incident response, root cause analysis

2023–current **Open source volunteer**, *20 %*

Reduced workload to 80 % in 2023 to dedicate time to maintaining and contributing to FOSS. Selected projects:

Lumen Labs
- Lead maintainer of the Lux package manager for Lua (written in Rust)

NixOS
- Maintainer of various packages and NixOS modules
- Member of the nixpkgs-committers, Neovim and Lua maintainer teams

Neovim
- Core contributions
- Maintainer of various popular plugins and GitHub actions workflows, including:
  - rustaceanvim (Rust)
  - haskell-tools.nvim
  - neotest-haskell
  - rocks.nvim
  - kickstart-nix.nvim
  - luarocks-tag-release

11/2017–01/2022 **Software engineer**, *Vela Solaris AG*

Java 17, Kotlin, Docker Swarm, Gradle

Polysun Simulation Software *(desktop app written in Java)*
- Simulation models: Batteries, PV/PVT, controllers, thermal components (heat pumps, storage tanks, co-generators, etc.), eMobility.
  Examples: Implementation of a new battery model;
  control algorithms for PV, batteries and heat pumps.
  Improvement of existing models.
- Plugins (e.g. for co-simulations with Python/Matlab/Simulink/PLC).
- Charting, reporting, UX.
- Publications, presentations and advanced workshops (e.g. at conferences).

Polysun BIM *(automation of engineering workflows in the building sector)*
- Conducted customer interviews & translated feedback into requirements.
- Rapid prototyping & concept validation with pilot customers.
- Roadmap planning.
- Software engineering/development:
  - *Desktop app with a modular bundle architecture*
    (Java, Kotlin, Hibernate (H2), Vavr, ReactiveX, Protobuf, ArchUnit).
  - *Cloud platform (backend), microservice architecture*
    (Spring Boot 2, Kafka Streams, MongoDB, Graphwalker).

Infrastructure

- CI/CD: Jenkins, Docker Swarm on AWS, e2e testing.
- Fully automated the Polysun release pipeline.
- Internal cloud infrastructure (e.g., licensing, sales automation, ...)
  Apache Camel.
- Developer tooling (CLI applications in Haskell, Python).

2013–2017 **Research assistant**, *PVprog, PVstore, TwinPower* , HTW Berlin
Development of forecast-based operational strategies for PV storage systems.
Optimisation of PV systems with batteries and heat pumps.
Implementation of simulation models in Matlab.

2011 **Factory worker**, Brose Fahrzeugteile, Leuwico, Lasco Umformtechnik

2010 **Math & english tutor**, Studentenring Coburg

2009 **Social service**, reha Gmbh

Education

2016–2017 **Master of Science - Renewable Energy Systems**,
Hochschule für Technik und Wirtschaft, HTW Berlin
GPA: A (with honours), HTW Berlin.

Thesis: Development of model-based control applications compliant with IEC 61499 for building energy systems with a focus on photovoltaics

Summary: Development of intelligent control applications for PV, battery and heat pump systems in compliance with IEC 61499. Development of communication interfaces for simulation tools such as Polysun and Matlab. Validation via co-simulations. Extension of the runtime environment (4diac-RTE) with a REST communication interface and set-up of a field test.

2012–2016 **Bachelor of Science - Renewable Energy Systems**,
Hochschule für Technik und Wirtschaft, HTW Berlin
GPA: A (with honours), HTW Berlin.

Thesis: Optimisation of grid feed-in from PV storage systems distributed across Germany using forecast-based operating strategies

Summary: Modelling and highly parallel CUDA simulation of 46126 PV storage systems distributed across Germany. Examination of the cumulated influence of various operational strategies and optimisation of forecasting- and control algorithms regarding self-sufficiency and grid integration.

Interests

Professional: FOSS/Linux development, trade literature, (un)conferences, hackathons, ensemble programming

Leisure: Cooking, urban gardening, running, cycling, hiking

Personal

Date of birth: 1989/04/18
Nationality: United Kingdom + Germany
Swiss residence: C permit

Languages

Native: English + German
B1: French

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